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## Chapter 17 Fatty Acid Catabolism

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### Multiple Choice Questions

1. Lipoprotein lipase acts in:

- A) hydrolysis of triacylglycerols of plasma lipoproteins to supply fatty acids to various tissues.
- B) intestinal uptake of dietary fat.
- C) intracellular lipid breakdown of lipoproteins.
- D) lipoprotein breakdown to supply needed amino acids.
- E) none of the above.

**Digestion, mobilization, and transport of fats**

**Pages: 668–669    Difficulty: 2    Ans: A**

2. Free fatty acids in the bloodstream are:

- A) bound to hemoglobin.
- B) carried by the protein serum albumin.
- C) freely soluble in the aqueous phase of the blood.
- D) nonexistent; the blood does not contain free fatty acids.
- E) present at levels that are independent of epinephrine.

**Digestion, mobilization, and transport of fats**

**Pages: 669–670    Difficulty: 2    Ans: B**

3. The role of hormone-sensitive triacylglycerol lipase is to:

- A) hydrolyze lipids stored in the liver.
- B) hydrolyze membrane phospholipids in hormone-producing cells.
- C) hydrolyze triacylglycerols stored in adipose tissue.
- D) synthesize lipids in adipose tissue.
- E) synthesize triacylglycerols in the liver.

**Digestion, mobilization, and transport of fats**

**Pages: 669–670    Difficulty: 2    Ans: C**

4. The glycerol produced from the hydrolysis of triacylglycerides enters glycolysis as:

- A) glucose.
- B) glucose-6-phosphate.
- C) dihydroxyacetone phosphate.
- D) pyruvate.
- E) glyceryl CoA.

**Digestion, mobilization, and transport of fats**

**Pages: 671    Difficulty: 2    Ans: C**

5. Transport of fatty acids from the cytoplasm to the mitochondrial matrix requires:

- A) ATP, carnitine, and coenzyme A.
- B) ATP, carnitine, and pyruvate dehydrogenase.
- C) ATP, coenzyme A, and hexokinase.
- D) ATP, coenzyme A, and pyruvate dehydrogenase.
- E) carnitine, coenzyme A, and hexokinase.

**Digestion, mobilization, and transport of fats**

**Pages: 671–672    Difficulty: 1    Ans: A**

6. Fatty acids are activated to acyl-CoAs and the acyl group is further transferred to carnitine because:

- A) acyl-carnitines readily cross the mitochondrial inner membrane, but acyl-CoAs do not.
- B) acyl-CoAs easily cross the mitochondrial membrane, but the fatty acids themselves will not.
- C) carnitine is required to oxidize  $\text{NAD}^+$  to NADH.
- D) fatty acids cannot be oxidized by FAD unless they are in the acyl-carnitine form.
- E) None of the above is true.

**Digestion, mobilization, and transport of fats**

**Pages: 671–672    Difficulty: 2    Ans: A**

7. Carnitine is:

- A) a 15-carbon fatty acid.
- B) an essential cofactor for the citric acid cycle.
- C) essential for intracellular transport of fatty acids.
- D) one of the amino acids commonly found in protein.
- E) present only in carnivorous animals.

**Digestion, mobilization, and transport of fats**

**Pages: 671–672    Difficulty: 2    Ans: C**

8. Which of these is able to cross the inner mitochondrial membrane?

- A) Acetyl-CoA
- B) Fatty acyl-carnitine
- C) Fatty acyl-CoA
- D) Malonyl-CoA
- E) None of the above can cross.

**Digestion, mobilization, and transport of fats**

**Pages: 671–672    Difficulty: 2    Ans: B**

9. What is the correct order of function of the following enzymes of  $\beta$ -oxidation?

1.  $\beta$ -Hydroxyacyl-CoA dehydrogenase
2. Thiolase
3. Enoyl-CoA hydratase
4. Acyl-CoA dehydrogenase

- A) 1, 2, 3, 4
- B) 3, 1, 4, 2
- C) 4, 3, 1, 2
- D) 1, 4, 3, 2
- E) 4, 2, 3, 1

**Oxidation of fatty acids****Pages: 673–674    Difficulty: 2    Ans: C**

10. If the 16-carbon saturated fatty acid palmitate is oxidized completely to carbon dioxide and water (via the  $\beta$ -oxidation pathway and the citric acid cycle), and all of the energy-conserving products are used to drive ATP synthesis in the mitochondrion, the net yield of ATP per molecule of palmitate is:

- A) 3.
- B) 10.
- C) 25.
- D) 108.
- E) 1000.

**Oxidation of fatty acids****Pages: 674–676    Difficulty: 2    Ans: D**

11. Saturated fatty acids are degraded by the stepwise reactions of  $\beta$ -oxidation, producing acetyl-CoA. Under aerobic conditions, how many ATP molecules would be produced as a consequence of removal of each acetyl-CoA?

- A) 2
- B) 3
- C) 4
- D) 5
- E) 6

**Oxidation of fatty acids****Pages: 674–676    Difficulty: 2    Ans: C**

12. Which of the following is (are) *true* of the oxidation of 1 mol of palmitate (a 16-carbon saturated fatty acid; 16:0) by the  $\beta$ -oxidation pathway, beginning with the free fatty acid in the cytoplasm?

- 1. Activation of the free fatty acid requires the equivalent of two ATPs.
- 2. Inorganic pyrophosphate ( $\text{PP}_i$ ) is produced.
- 3. Carnitine functions as an electron acceptor.
- 4. 8 mol of  $\text{FADH}_2$  are formed.
- 5. 8 mol of acetyl-CoA are formed.
- 6. There is no direct involvement of  $\text{NAD}^+$ .

- A) 1 and 5 only
- B) 1, 2, and 5
- C) 1, 2, and 6
- D) 1, 3, and 5
- E) 5 only

**Oxidation of fatty acids****Pages: 673–676    Difficulty: 2    Ans: B**

13. Complete oxidation of 1 mole of which fatty acid would yield the most ATP?

- A) 16-carbon saturated fatty acid
- B) 18-carbon mono-unsaturated fatty acid
- C) 16-carbon mono-unsaturated fatty acid
- D) 16-carbon poly-unsaturated fatty acid
- E) 14-carbon saturated fatty acid

**Oxidation of fatty acids****Pages: 673–678    Difficulty: 2    Ans: B**

14. Which of the following statements apply (applies) to the  $\beta$ -oxidation of fatty acids?

- 1. The process takes place in the cytosol of mammalian cells.
- 2. Carbon atoms are removed from the acyl chain one at a time.
- 3. Before oxidation, fatty acids must be converted to their CoA derivatives.
- 4.  $\text{NADP}^+$  is the electron acceptor.
- 5. The products of  $\beta$ -oxidation can directly enter the citric acid cycle for further oxidation.

- A) 1 and 3 only
- B) 1, 2, and 3
- C) 1, 2, and 5
- D) 3 and 5 only
- E) 4 only

**Oxidation of fatty acids****Pages: 673–676    Difficulty: 1    Ans: D**

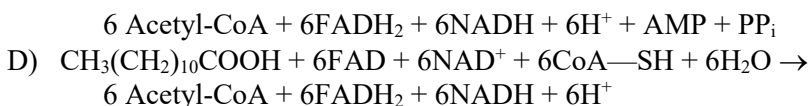
15. Which of the following statements concerning the  $\beta$ -oxidation of fatty acids is *true*?

- A) About 1200 ATP molecules are ultimately produced per 20-carbon fatty acid oxidized.
- B) One  $\text{FADH}_2$  and two  $\text{NADH}$  are produced for each acetyl-CoA.
- C) The free fatty acid must be carboxylated in the  $\beta$  position by a biotin-dependent reaction before the process of  $\beta$ -oxidation commences.
- D) The free fatty acid must be converted to a thioester before the process of  $\beta$ -oxidation commences.
- E) Two  $\text{NADH}$  are produced for each acetyl-CoA.

**Oxidation of fatty acids****Pages: 673–676    Difficulty: 3    Ans: D**

16. The balanced equation for the degradation of  $\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$  via the  $\beta$ -oxidation pathway is:

- A)  $\text{CH}_3(\text{CH}_2)_{10}\text{COOH} + 5\text{FAD} + 5\text{NAD}^+ + 6\text{CoA-SH} + 5\text{H}_2\text{O} + \text{ATP} \rightarrow$   
 $6 \text{ Acetyl-CoA} + 5\text{FADH}_2 + 5\text{NADH} + 5\text{H}^+ + \text{AMP} + \text{PP}_i$
- B)  $\text{CH}_3(\text{CH}_2)_{10}\text{COOH} + 5\text{FAD} + 5\text{NAD}^+ + 6\text{CoA-SH} + 5\text{H}_2\text{O} \rightarrow$   
 $6 \text{ Acetyl-CoA} + 5\text{FADH}_2 + 5\text{NADH} + 5\text{H}^+$
- C)  $\text{CH}_3(\text{CH}_2)_{10}\text{COOH} + 6\text{FAD} + 6\text{NAD}^+ + 6\text{CoA-SH} + 6\text{H}_2\text{O} + \text{ATP} \rightarrow$

**Oxidation of fatty acids****Pages: 673–676    Difficulty: 3    Ans: A**17. Which compound is an intermediate of the  $\beta$ -oxidation of fatty acids?

- A)  $\text{CH}_3-(\text{CH}_2)_{20}-\text{CO}-\text{COOH}$   
 B)  $\text{CH}_3-\text{CH}_2-\text{CO}-\text{CH}_2-\text{CO}-\text{OPO}_3^{2-}$   
 C)  $\text{CH}_3-\text{CH}_2-\text{CO}-\text{CH}_2-\text{OH}$   
 D)  $\text{CH}_3-\text{CH}_2-\text{CO}-\text{CO}-\text{S}-\text{CoA}$   
 E)  $\text{CH}_3-\text{CO}-\text{CH}_2-\text{CO}-\text{S}-\text{CoA}$

**Oxidation of fatty acids****Page: 673    Difficulty: 1    Ans: E**18. The conversion of palmitoyl-CoA (16:0) to myristoyl-CoA (14:0) and 1 mol of acetyl-CoA by the  $\beta$ -oxidation pathway results in the net formation of:

- A) 1  $\text{FADH}_2$  and 1  $\text{NADH}$ .  
 B) 1  $\text{FADH}_2$  and 1  $\text{NADPH}$ .  
 C) 1  $\text{FADH}_2$ , 1  $\text{NADH}$ , and 1  $\text{ATP}$ .  
 D) 2  $\text{FADH}_2$  and 2  $\text{NADH}$ .  
 E) 2  $\text{FADH}_2$ , 2  $\text{NADH}$ , and 1  $\text{ATP}$ .

**Oxidation of fatty acids****Page: 673    Difficulty: 2    Ans: A**19. Which of the following is *not* true regarding the oxidation of 1 mol of palmitate (16:0) by the  $\beta$ -oxidation pathway?

- A) 1 mol of  $\text{ATP}$  is needed.  
 B) 8 mol of acetyl-CoA are formed.  
 C) 8 mol of  $\text{FADH}_2$  are formed.  
 D)  $\text{AMP}$  and  $\text{PP}_i$  are formed.  
 E) The reactions occur in the mitochondria.

**Oxidation of fatty acids****Pages: 673–676    Difficulty: 2    Ans: C**20. If an aerobic organism (e.g., the bacterium *E. coli*) were fed each of the following four compounds as a source of energy, the energy yield per mole from these molecules would be in the order:

- A) alanine > glucose > palmitate (16:0)  
 B) glucose > alanine > palmitate  
 C) glucose > palmitate > alanine  
 D) palmitate > alanine > glucose

E) palmitate > glucose > alanine

**Oxidation of fatty acids**

**Pages: 675–676    Difficulty: 3    Ans: E**

21. Which of the following is (are) *true* of the  $\beta$ -oxidation of long-chain fatty acids?

1. The enzyme complex that catalyzes the reaction contains biotin.
2.  $\text{FADH}_2$  serves as an electron carrier.
3. NADH serves as an electron carrier.
4. Oxidation of an 18-carbon fatty acid produces six molecules of propionyl-CoA.
5. Oxidation of a 15-carbon fatty acid produces at least one propionyl-CoA.

- A) 1, 2, and 3  
B) 1, 2, and 5  
C) 2, 3, and 4  
D) 2, 3, and 5  
E) 3 and 5 only

**Oxidation of fatty acids**

**Pages: 673–676    Difficulty: 2    Ans: D**

22. The following fatty acid, in which the indicated carbon is labeled with  $^{14}\text{C}$ , is fed to an animal:



After allowing 30 minutes for fatty acid  $\beta$ -oxidation, the label would most likely be recovered in:

- A) acetyl-CoA.  
B) beta-hydroxy butyryl-CoA.  
C) both acetyl-CoA and propionyl-CoA.  
D) palmitoyl-CoA.  
E) propionyl-CoA.

**Oxidation of fatty acids**

**Pages: 677–678    Difficulty: 2    Ans: E**

23. The carbon atoms from a fatty acid with an odd number of carbons will enter the citric acid cycle as acetyl-CoA and:

- A) butyrate.  
B) citrate.  
C) malate.  
D) succinyl-CoA.  
E)  $\alpha$ -ketoglutarate.

**Oxidation of fatty acids**

**Pages: 677–678    Difficulty: 2    Ans: D**

24. In the disease sprue, vitamin B<sub>12</sub> (cobalamin) is poorly absorbed in the intestine, resulting in B<sub>12</sub> deficiency. If each of the following fatty acids were in the diet, for which one would the process of fatty acid oxidation be *most* affected in a patient with sprue?

- A) CH<sub>3</sub>(CH<sub>2</sub>)<sub>10</sub>COOH
- B) CH<sub>3</sub>(CH<sub>2</sub>)<sub>11</sub>COOH
- C) CH<sub>3</sub>(CH<sub>2</sub>)<sub>12</sub>COOH
- D) CH<sub>3</sub>(CH<sub>2</sub>)<sub>14</sub>COOH
- E) CH<sub>3</sub>(CH<sub>2</sub>)<sub>18</sub>COOH

**Oxidation of fatty acids**

**Pages: 677–678    Difficulty: 2    Ans: B**

25. Which enzyme is the major regulatory control point for β-oxidation?

- A) Pyruvate carboxylase
- B) Carnitine acyl transferase I
- C) Acetyl CoA dehydrogenase
- D) Enoyl CoA isomerase
- E) Methylmalonyl CoA mutase

**Oxidation of fatty acids**

**Page: 679    Difficulty: 2    Ans: B**

26. The metabolite that regulates the activity of carnitine acyl transferase I is:

- A) acetyl-CoA.
- B) carnitine-CoA.
- C) malonyl-CoA.
- D) NADH.
- E) CoA.

**Oxidation of fatty acids**

**Page: 679    Difficulty: 2    Ans: C**

27. Which of the following is *not* true about Vitamin B<sub>12</sub>?

- A) Vitamin B<sub>12</sub> contains a porphyrin ring.
- B) Vitamin B<sub>12</sub> contains a carbon-cobalt bond.
- C) Vitamin B<sub>12</sub> readily undergoes homolytic bond cleavage.
- D) Deficiency of Vitamin B<sub>12</sub> causes pernicious anemia.
- E) Vitamin B<sub>12</sub> catalyzes hydrogen atom exchange with solvent H<sub>2</sub>O.

**Oxidation of fatty acids**

**Pages: 680–681    Difficulty: 2    Ans: E**

28. During β-oxidation of fatty acids, \_\_\_\_\_ is produced in peroxisomes but not in mitochondria.

- A) acetyl-CoA

- B)  $\text{FADH}_2$
- C)  $\text{H}_2\text{O}$
- D)  $\text{H}_2\text{O}_2$
- E) NADH

**Oxidation of fatty acids****Page: 682 Difficulty: 2 Ans: D**

29. When comparing the  $\beta$ -oxidation and  $\omega$ -oxidation pathways, which one of the following statements is *correct*?

- A)  $\beta$ -oxidation and  $\omega$ -oxidation occur in the cytoplasm.
- B)  $\beta$ -oxidation occurs at the carboxyl end of the fatty acid whereas  $\omega$  oxidation occurs at the methyl end.
- C)  $\beta$ -oxidation occurs at the methyl end of the fatty acid, whereas  $\omega$ -oxidation occurs at the carboxyl end.
- D)  $\beta$ -oxidation occurs mainly in the cytoplasm, whereas  $\omega$ -oxidation occurs mainly in the mitochondria.
- E)  $\beta$ -oxidation occurs mainly in the mitochondria, whereas  $\omega$ -oxidation occurs mainly in the cytoplasm.

**Oxidation of fatty acids****Pages: 684–685 Difficulty: 2 Ans: B**

30. Ketone bodies are formed in the liver and transported to the extrahepatic tissues mainly as:

- A) acetoacetyl-CoA.
- B) acetone.
- C)  $\beta$ -hydroxybutyric acid.
- D)  $\beta$ -hydroxybutyryl-CoA.
- E) lactic acid.

**Ketone bodies****Pages: 686–687 Difficulty: 2 Ans: C**

31. The major site of formation of acetoacetate from fatty acids is the:

- A) adipose tissue.
- B) intestinal mucosa.
- C) kidney.
- D) liver.
- E) muscle.

**Ketone bodies****Pages: 686–687 Difficulty: 1 Ans: D**

## Short Answer Questions

32. Why is it more efficient to store energy as lipid rather than as glycogen?

**Ans:** First, the energy yield per gram of lipid (about 38 kJ/g) is more than twice that for carbohydrate (about 17 kJ/g). Second, lipid is stored as anhydrous lipid droplets, but carbohydrates such as glycogen and starch are stored hydrated, and the water of hydration roughly triples the effective weight of the carbohydrate, reducing the energy yield to about 6 kJ/g.

### Digestion, mobilization, and transport of fats

Page: 667 Difficulty: 1

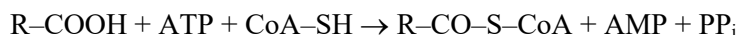
33. Explain why lipases are required in *both* the intestine and in the bloodstream?

**Ans:** Fats ingested in the diet are mostly found as triacylglycerols. In order to be taken up through the intestinal mucosa these triacylglycerols must be converted to free fatty acids by the intestinal lipases. Once in the bloodstream, the free fatty acids are again converted to triacylglycerols for transport in chylomicrons. For absorption into remote tissues from the bloodstream, the triacylglycerols must again be converted to free fatty acids, this time by lipases found in the bloodstream.

### Digestion, mobilization, and transport of fats

Page: 668 Difficulty: 2

34. In the first step of fatty acid oxidation, the fatty acid (R—COOH) is converted to its coenzyme A derivative in the following reaction:



The standard free-energy change ( $\Delta G^\circ$ ) for this reaction is  $-15 \text{ kJ/mol}$

What will tend to make the reaction even more favorable when it takes place within a cell?

**Ans:** The hydrolysis of  $\text{PP}_i$  by inorganic pyrophosphatase, for which  $\Delta G^\circ$  is  $-19 \text{ kJ/mol}$ , makes the overall  $\Delta G^\circ$  more negative.

### Digestion, mobilization, and transport of fats

Page: 671 Difficulty: 2

35. The oxidation of acetyl-CoA added to isolated, intact mitochondria is stimulated strongly by carnitine. Why?

**Ans:** Carnitine is essential in the transport of fatty acyl groups into the mitochondrial matrix, where fatty acid oxidation occurs.

### Digestion, mobilization, and transport of fats

Pages: 671–672 Difficulty: 1

36. Draw the first two steps of  $\beta$ -oxidation of saturated fatty acids. Show structures and indicate where any cofactors participate.

**Ans:** The reactions are those catalyzed by fatty acyl-CoA dehydrogenase and enoyl hydratase. Note

the formation of  $\text{FADH}_2$  for the first step and the addition of water across the double bond in the second step. (See Fig. 17-8a.)

**Oxidation of fatty acids**

**Page: 673 Difficulty: 2**

37. Draw the third and fourth steps in the  $\beta$ -oxidation of a saturated fatty acid. Show structures, name enzymes, and indicate where any cofactors participate.

**Ans:** The reactions are those catalyzed by  $\beta$ -hydroxyl-CoA dehydrogenase and thiolase. Note the formation of NADH for the third step and the cleavage of the carbon-carbon bond in a reverse Claisen condensation for the fourth step. See Fig. 17-8a.

**Oxidation of fatty acids**

**Page: 673 Difficulty: 3**

38. One of the steps in fatty acid oxidation in mitochondria involves the addition of water across a double bond. What is the next step in the process? Show structures and indicate where any cofactor(s) participate(s).

**Ans:** The reaction is that catalyzed by  $\beta$ -hydroxyacyl-CoA dehydrogenase, for which  $\text{NAD}^+$  is cofactor. (See Fig. 17-8a.)

**Oxidation of fatty acids**

**Page: 673 Difficulty: 3**

39. Explain the chemical reason why the addition of water across the double bond in the second step of  $\beta$ -oxidation occurs with the OH adding at the  $\beta$ -carbon, not the  $\alpha$ -carbon. Also, how does addition at the  $\beta$ -carbon aid in the subsequent step(s) of  $\beta$ -oxidation?

**Ans:** Recall that the carbon that is  $\alpha$  to a carbonyl is readily able to become a carbanion; i.e. it is not a good electrophilic center for nucleophilic addition of a hydroxide (see Chapter 13). The  $\beta$ -carbon, by contrast is a good electrophilic center as can be readily shown by “pushing” the electrons from the  $\text{C}=\text{C}$  double bond onto the carbonyl oxygen. Addition of the OH in the  $\beta$ -position is useful for later steps since after oxidation of the OH to a carbonyl, the  $\alpha$ -carbon is even more susceptible to becoming a carbanion, as is needed for the reverse Claisen condensation catalyzed by thiolase.

**Oxidation of fatty acids**

**Page: 673 Difficulty: 3**

40. In the citric acid cycle, a double bond is introduced into a four-carbon compound containing the  $-\text{CH}_2-\text{CH}_2-$  group, producing fumarate. Show a similar reaction that occurs in the  $\beta$ -oxidation pathway.

**Ans:** This is the reaction catalyzed by acyl-CoA dehydrogenase. See Fig. 17-8a.

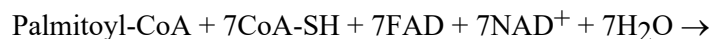
**Oxidation of fatty acids**

**Page: 673 Difficulty: 2**

41. Write a balanced equation for the  $\beta$ -oxidation of palmitoyl-CoA, a 16-carbon, fully saturated fatty acid,

and indicate how much of each product is formed.

**Ans:** The overall reaction is:



### Oxidation of fatty acids

**Pages: 673–675      Difficulty: 3**

42. For each two-carbon increase in the length of a saturated fatty acid chain, how many additional moles of ATP can be formed upon complete oxidation of one mole of the fatty acid to  $\text{CO}_2$  and  $\text{H}_2\text{O}$ ?

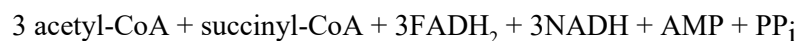
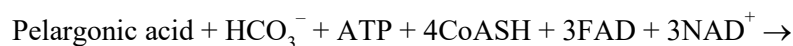
**Ans:** Each  $-\text{CH}_2-\text{CH}_2-$  unit yields 14 extra ATP molecules. The two oxidations of the  $\beta$ -oxidation pathway produce 1  $\text{FADH}_2$  and 1  $\text{NADH}$ , which yield 1.5 and 2.5 ATP, respectively, by oxidative phosphorylation. The extra acetyl-CoA, when oxidized via the citric acid cycle, yields another 10 ATP equivalents: 3  $\text{NADH}$ , 1  $\text{FADH}_2$ , and 1 ATP or GTP.

### Oxidation of fatty acids

**Pages: 673–676      Difficulty: 3**

43. Write a balanced equation for the complete oxidation (to acetyl-CoA and any other products that might be formed) of pelargonic acid,  $\text{CH}_3(\text{CH}_2)_7\text{COOH}$ .

**Ans:** The odd-chain fatty acid is first activated to the CoA derivative, then oxidized to 3 acetyl-CoA and 1 propionyl-CoA by  $\beta$ -oxidation. The propionyl-CoA is converted to succinyl-CoA through the sequence of reactions shown in Fig. 17-12. The overall reaction is therefore:

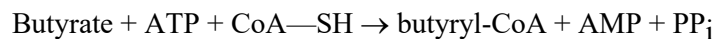


### Oxidation of fatty acids

**Pages: 673–678      Difficulty: 3**

44. Describe the steps in the metabolic pathway in which cells oxidize a four-carbon, straight-chain, saturated fatty acid (butyrate; 4:0) to the fragments that enter the citric acid cycle. Show the structures of intermediates and products, and indicate where any cofactor(s) participate(s).

**Ans:** Butyrate is first activated:



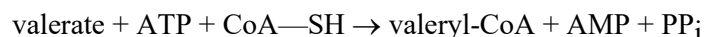
Then, the butyryl group is transferred to carnitine and transported into the mitochondrial matrix, where it is reconverted to the butyryl-CoA derivative. This passes through the four steps of  $\beta$ -oxidation. (See Fig. 17-8a.)

### Oxidation of fatty acids

**Pages: 673–676****Difficulty: 3**

45. Describe the steps in the metabolic pathway in which cells oxidize a five-carbon, straight-chain, saturated fatty acid (valerate; 5:0) to the fragments that enter the citric acid cycle. Show the structures of intermediates and products, and indicate where any cofactor(s) participate(s).

**Ans:** Valerate is first activated:



Then, the valeryl group is transferred to carnitine and transported into the mitochondrial matrix, where it is reconverted to the valeryl;-CoA derivative. This passes through the four steps of  $\beta$ -oxidation producing acetyl-CoA and propionyl-CoA. (See Fig. 17-8a.) Propionyl-CoA would be converted to succinyl-CoA by the reaction sequence in Fig. 17-12.

**Oxidation of fatty acids****Pages: 673–676****Difficulty: 3**

46. An experimenter studying the oxidation of fatty acids in extracts of liver found that when palmitate (16:0) was provided as substrate, it was completely oxidized to  $\text{CO}_2$ . However, when undecanoic acid (11:0) was added as substrate, incomplete oxidation occurred unless he bubbled  $\text{CO}_2$  through the reaction mixture. The addition of the protein avidin, which binds tightly to biotin, prevented the complete oxidation of undecanoic acid even in the presence of  $\text{CO}_2$ , although it had no effect on palmitate oxidation. Explain these observations in light of what you know of fatty acid oxidation reactions.

**Ans:** Oxidation of odd-chain fatty acid yields acetyl-CoA + propionyl-CoA. The reaction  $\text{CO}_2 + \text{propionyl-CoA} \rightarrow \text{methylmalonyl-CoA}$  is catalyzed by propionyl-CoA carboxylase, a biotin-containing enzyme, which is therefore inhibited by avidin.

**Oxidation of fatty acids****Pages: 677–678****Difficulty: 3**

47. Two vitamins, biotin and vitamin  $\text{B}_{12}$ , play crucial roles in the metabolism of propionic acid (propionate). Explain this by showing the steps in which each is essential in propionate metabolism.

**Ans:** Biotin and vitamin  $\text{B}_{12}$  act as cofactors for propionyl-CoA carboxylase and methylmalonyl-CoA mutase, respectively; see Fig. 17-12 for the complete sequence of reactions.

**Oxidation of fatty acids****Page: 678 Difficulty: 3**

48. The total degradation of a fatty acid with an odd number of carbons yields acetyl-CoA and another compound, X. Show the structure of X and describe the pathway by which it is converted into a citric acid cycle intermediate, including where any cofactors participate.

**Ans:** X is propionyl-CoA, and its conversion into succinyl-CoA is accomplished by the reactions in Fig. 17-12.

**Oxidation of fatty acids**

**Page: 678 Difficulty: 3**

49. What are the two major differences between the  $\beta$ -oxidation pathway in mitochondria and the  $\beta$ -oxidation pathway in peroxisomes?

**Ans:** The first major difference lies in the detailed chemistry of the first of the four steps of  $\beta$ -oxidation. In the mitochondria, the reduced flavin that is a product of the first step is re-oxidized via the electron transport chain. In the peroxisomes, the reduced flavin is re-oxidized by  $O_2$ , thus producing  $H_2O_2$ . The second major difference is that the peroxisomal enzymes are much more active on longer (i.e., 26-carbons) fatty acids than the mitochondrial enzymes.

**Oxidation of fatty acids****Page: 682 Difficulty: 3**

50. Briefly explain the basic concept of  $\omega$ -oxidation.

**Ans:** Relatively short chain fatty acids (i.e., 10-12 carbons) are oxidized on the  $\omega$ -end (in three steps) to form a second carboxyl group. The “double-ended” fatty acid is then taken into the mitochondria and subjected to normal  $\beta$ -oxidation, except that the last product is not acetyl-CoA, but rather succinate or adipate.

**Oxidation of fatty acids****Pages: 684–685 Difficulty: 2**

51. If you received a laboratory report showing the presence of a high concentration of ketone bodies in the urine of a patient, what disease would you suspect? Why do ketone bodies accumulate in such patients?

**Ans:** The patient is probably an untreated diabetic, but the condition might also result from fasting. In either case, the unavailability of glucose from the blood stimulates gluconeogenesis in the liver. As the substrate for glucose formation, oxaloacetate is withdrawn from the citric acid cycle, bringing that cycle to a near halt. The fatty acids being oxidized in the liver yield acetyl-CoA, which now cannot be oxidized via the citric acid cycle. Reversal of the thiolase reaction produces acetoacetyl-CoA, which is then converted into ketone bodies and exported from the liver. (See Fig. 17-18, p. 666.)

**Ketone bodies****Pages: 686–688 Difficulty: 2**

52. Draw the structure of one ketone body, and describe circumstances under which you would expect to find high concentrations of this compound in the urine of a human.

**Ans:** The ketone bodies, acetoacetate,  $\beta$ -hydroxybutyrate, and acetone (Fig. 17-19), are overproduced in untreated diabetes mellitus and during prolonged fasting, when fatty acids become the principle energy source.

**Ketone bodies****Pages: 686–688 Difficulty: 2**